

Headwater

AP Calculus AB

Ashfell Dam from the inside — galleries, a gate chamber, and a hundred metres of air off the crest.

45	8	5.2	15/15	0	9 · 32 · 4
STOPS	CHOICE (18%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Nothing is wrong with this campaign. All fifteen equations computed, no debt, **no select-only concepts at all**, no absent concepts, and four OPERATE stops — the most in the set. It already carries SWEEP, PROBE, HOLDOUT, RESIDUAL and PROPAGATE. It is the worked example of what every other plan in this set is aiming at, and it is included here only because the question was asked of every high-school campaign.

Everything below is therefore a **fun** argument and nothing else. There is no curriculum case for any of it, and if the schedule is short this is the campaign to skip.

PROTECTED — DO NOT CONVERT

All thirteen DERIVE stops stay. Headwater is AP Calculus AB and the derivation rail is the course. Nothing below touches one.

NOW The mix as it ships

DERIVE 13	BALLPARK 12	CHOICE 8	PROTOCOL 1	CHAIN 1	BALANCE 1	PROBE 1	SWEEP 1	DEGENERACY 1	PROPAGATE 1	VERIFY 1	HOLDOUT 1	TRIGGER 1	RESIDUAL 1	SCIENCETANK 1
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CURRICULUM What the syllabus is not getting

No equation debt. Every equation the syllabus lists is computed by a question somewhere in the campaign.

TIER Concepts the player only ever picks from a list

None. Every mechanism concept this campaign touches is reached at least once by a stop that makes the player build or operate something.

FUN What this subject and this place can carry

HOLD

STRONG

a gate-chamber stop

The best HOLD available in the catalogue. Hold the reservoir inside a closing band while an inflow surge arrives and you work the gates — rate in against rate out, which is the fundamental theorem with a hundred metres of air off the crest if you get it wrong. The game is a dam; the format is a dam.

SERVES accumulation and rate of change, felt rather than differentiated

NEEDS the band closing from the hydrology; a do-nothing run must fail

TRIAL STRONG

the galleries

A five-storey tower in a gorge with galleries, a gate chamber and no ceiling over the hallways. An inspection route driven in the order the water goes, graded on order.

SERVES the interior as a system rather than a set of rooms

RISK world work; the theme brings its own world module

SPOT MEDIUM

a gate stop

A gate-operating instruction replaced mid-flood. Legitimate, and it teaches operating discipline rather than calculus — which is fine as one stop and would be wrong as three.

TALLY MEDIUM

an inflow stop

Inflow readings accumulate into a defensible estimate. The game already carries RESIDUAL and PROPAGATE, so this competes with itself.

BELT REJECT

Nothing is sorted in a dam. Reject.

LOB REJECT

A spillway trajectory is the hook and it is computed in this course, on purpose, by the student. Reject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs headwater --snapshot before.json, then --against before.json .

Minimum honest 1 STOP · ONE SITTING

42	d14	GATES	CHOICE → HOLD	The only trace on the screen the only trace on the screen — held, while the surge arrives
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CHOICE 8 → 7 (16%) · moves 9·32·4 → 8·32·5 · effective formats 5.2 → 5.4 · 1 new operate stop, 1 of them arcade

Balanced 3 STOPS · THE RECOMMENDATION

42	d14	GATES	CHOICE → HOLD	The only trace on the screen
18	d6	SAFE	CHOICE → SPOT	The two peaks are not the same hour two peaks, under a gate instruction replaced mid-flood
22	d8	STORE	BALLPARK → TALLY	One number for six hours one number for six hours — accumulated until it is defensible

CHOICE 8 → 6 (13%) · moves 9·32·4 → 7·31·7 · effective formats 5.2 → 5.9 · 3 new operate stops, 2 of them arcade

Ambitious 4 STOPS · ADDS WORLD WORK

42	d14	GATES	CHOICE → HOLD	The only trace on the screen
18	d6	SAFE	CHOICE → SPOT	The two peaks are not the same hour
22	d8	STORE	BALLPARK → TALLY	One number for six hours
30	d10	STRUCT	CHOICE → TRIAL	What the correction does not touch the galleries, driven, in the order the water goes

CHOICE 8 → 5 (11%) · moves 9·32·4 → 6·31·8 · effective formats 5.2 → 6.1 · 4 new operate stops, 3 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline `curriculum-debt.json`. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent.

2. Snapshot. **3.** Edit the book, never the generated content — `tools/import-book.mjs`

`books/headwater.yml` `headwater --verify`. **4.** `npm run traps`: every instrument's trap must fire. **5.** `npm run check headwater` and `npm run drive headwater`, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** `--against` the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/headwater/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=headwater npm run dev`.